

Task 1: ML Classification Project

Alfido Tech Internship Program - Artificial Intelligence Track

Intern name	[Your name]
Task	Task 1 - ML Classification Project
Dataset	Breast Cancer Wisconsin Diagnostic (scikit-learn built-in, 569 samples, 30 features)
Final model	SVM (RBF) (RBF kernel)
GitHub repository	https://github.com//alfido-task1-ml-classification
Main notebook	https://github.com//alfido-task1-ml-classification/blob/main/notebooks/task1_ml_classification.ipynb
Environment	Python 3.12, scikit-learn 1.8.0, pandas 3.0.2, numpy 2.4.4, seed = 42

Summary

A supervised classifier was built to predict whether a breast tumour is malignant or benign. Four algorithms (Logistic Regression, Random Forest, SVM and Gradient Boosting) were compared with 5-fold stratified cross-validation and tuned with grid search. The final model, **SVM (RBF)**, reached **accuracy 0.974**, **precision 1.000**, **recall 0.929**, **F1 0.963** and **ROC-AUC 0.993** on a held-out test set of 114 samples (72 TN, 0 FP, 3 FN, 39 TP; positive class = malignant).

All four models scored within about one percentage point of each other, so the choice between the top models is not statistically decisive with a test set this small.

1. Problem and dataset

Goal: build and evaluate a supervised classification model. **Task:** predict tumour malignancy from 30 numeric features (radius, texture, perimeter, area, smoothness, concavity and so on) computed from digitised images of a fine-needle aspirate. Each feature appears as a mean, a standard error and a worst value.

The dataset has 569 samples: 357 benign (62.7%) and 212 malignant (37.3%). The imbalance is mild, so stratified splitting is used and F1 and recall are reported in addition to accuracy. The positive class is defined as **malignant**, so recall means the share of real cancers that were caught.

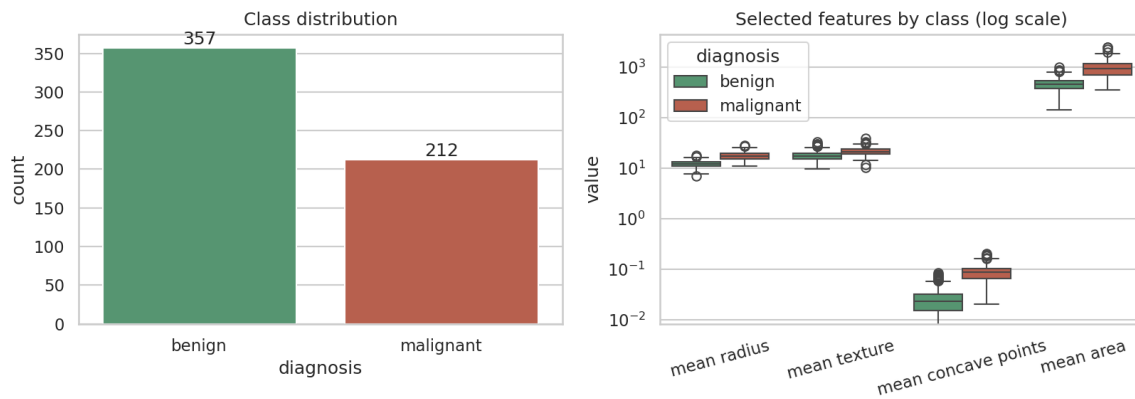


Figure 1. Class distribution and selected features by class (log scale).

2. Data preprocessing

- **Quality checks:** 0 missing values, 0 duplicate rows, all features numeric. Cleaning code (drop duplicates / missing) is included so the pipeline works on messier data.
- **Target encoding:** scikit-learn codes malignant as 0; it was flipped so malignant = 1 (positive class).
- **Scaling:** StandardScaler is placed *inside* the pipelines of Logistic Regression and SVM, so it is fitted on training folds only. This prevents data leakage. Tree models need no scaling.
- **Split:** 80/20 stratified train/test split (455 train, 114 test), random_state = 42. The test set is used once at the end.

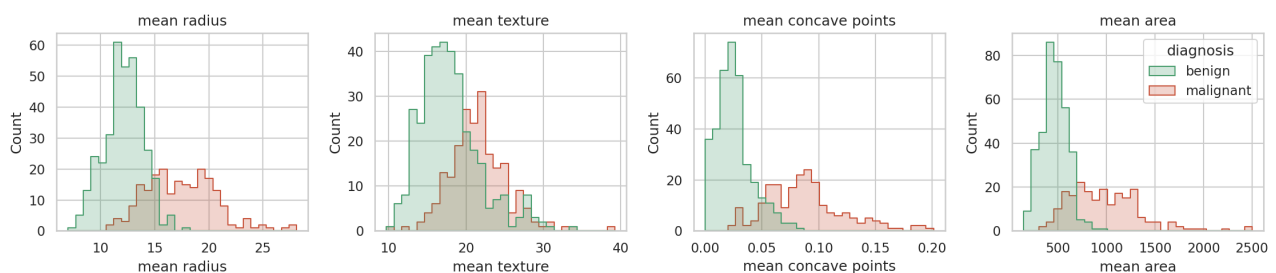


Figure 2. Feature distributions by class. Malignant tumours tend to have larger radius and area and more concave points.

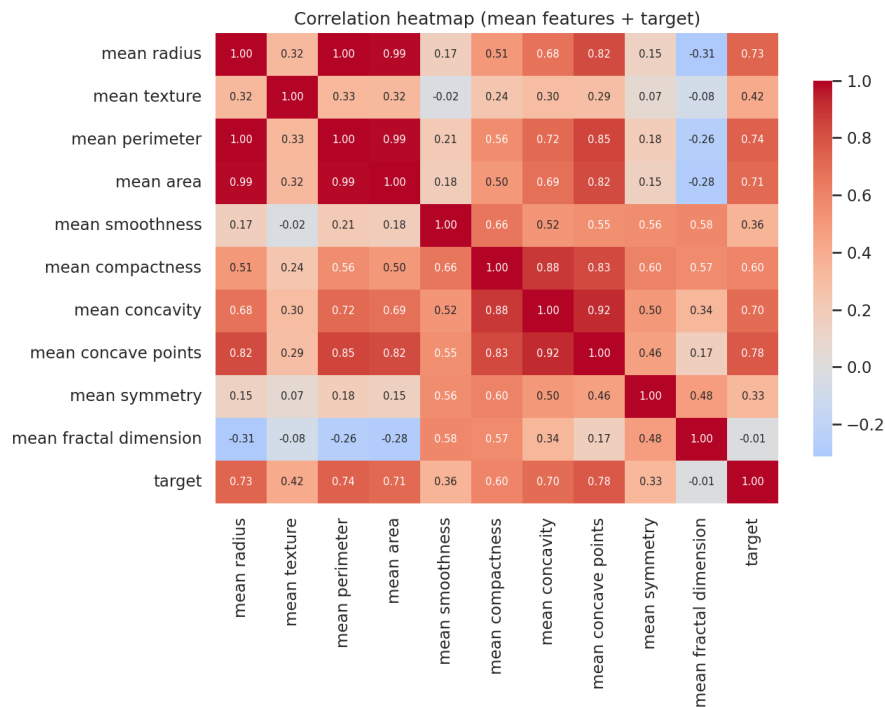


Figure 3. Correlation heatmap of the mean features and the target. Radius, perimeter and area are almost perfectly correlated.

3. Methodology

Cross-validation. Each algorithm was evaluated with 5-fold stratified cross-validation on the training set using accuracy, precision, recall, F1 and ROC-AUC.

Algorithms compared. Logistic Regression (linear baseline, interpretable), Random Forest (bagged trees), SVM with RBF kernel (non-linear margin classifier) and Gradient Boosting (boosted trees).

Tuning. GridSearchCV with 5-fold CV, optimising F1, on the training set only. Grids: Logistic Regression C; Random Forest n_estimators, max_depth, min_samples_leaf; SVM C and gamma; Gradient Boosting n_estimators, learning_rate, max_depth.

Selection rule. The final model is the one with the best cross-validated F1 on the training data, so the test set remains an unbiased estimate.

4. Results

4.1 Cross-validation (training set, before tuning), mean +/- std over 5 folds

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.974 +/- 0.015	0.977 +/- 0.028	0.953 +/- 0.040	0.964 +/- 0.021	0.996 +/- 0.005
Random Forest	0.960 +/- 0.016	0.958 +/- 0.022	0.935 +/- 0.039	0.946 +/- 0.023	0.988 +/- 0.007
SVM (RBF)	0.971 +/- 0.005	0.977 +/- 0.021	0.947 +/- 0.034	0.961 +/- 0.008	0.995 +/- 0.005
Gradient Boosting	0.967 +/- 0.014	0.965 +/- 0.021	0.947 +/- 0.034	0.955 +/- 0.020	0.991 +/- 0.006

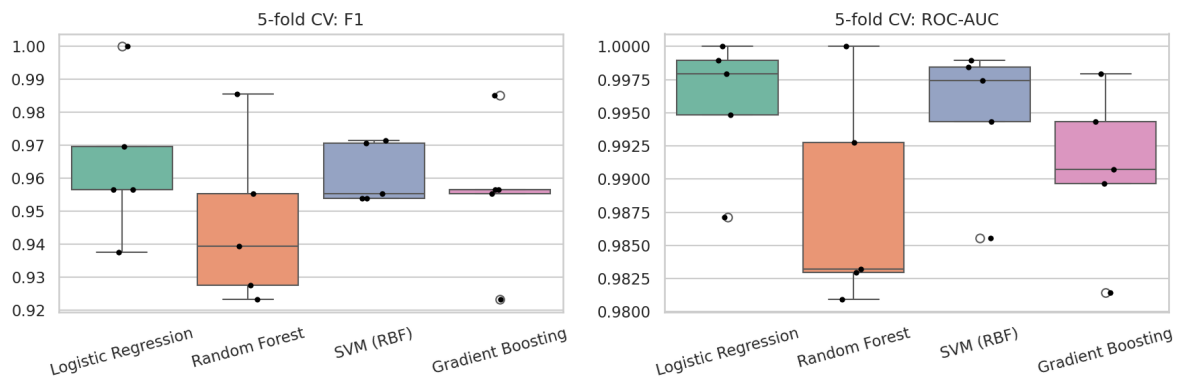


Figure 4. Distribution of per-fold F1 and ROC-AUC for the four algorithms.

4.2 Hyper-parameter tuning

Model	Best CV F1	Best params
Logistic Regression	0.964	{'C': 1}
Random Forest	0.9516	{'max_depth': 6, 'min_samples_leaf': 1, 'n_estimators': 200}
SVM (RBF)	0.9674	{'C': 10, 'gamma': 'scale'}
Gradient Boosting	0.9645	{'learning_rate': 0.1, 'max_depth': 2, 'n_estimators': 200}

4.3 Held-out test set (114 samples, tuned models)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.965	0.975	0.929	0.951	0.996
Random Forest	0.965	1.000	0.905	0.950	0.995
SVM (RBF)	0.974	1.000	0.929	0.963	0.993
Gradient Boosting	0.965	1.000	0.905	0.950	0.993

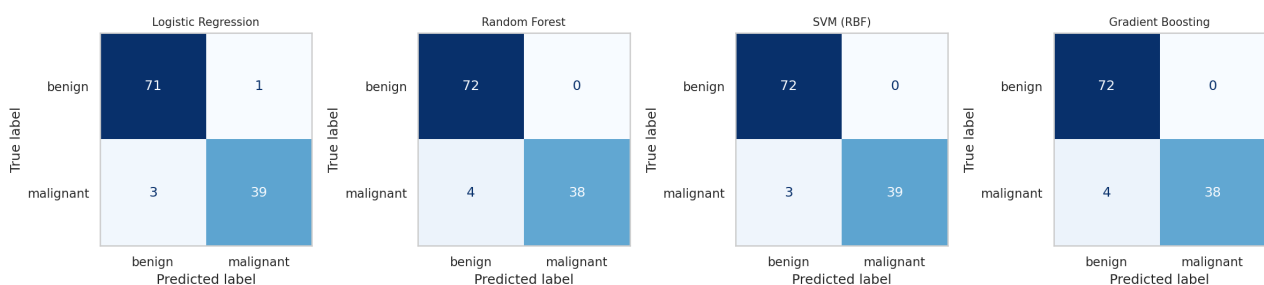


Figure 5. Test-set confusion matrices (positive class = malignant).

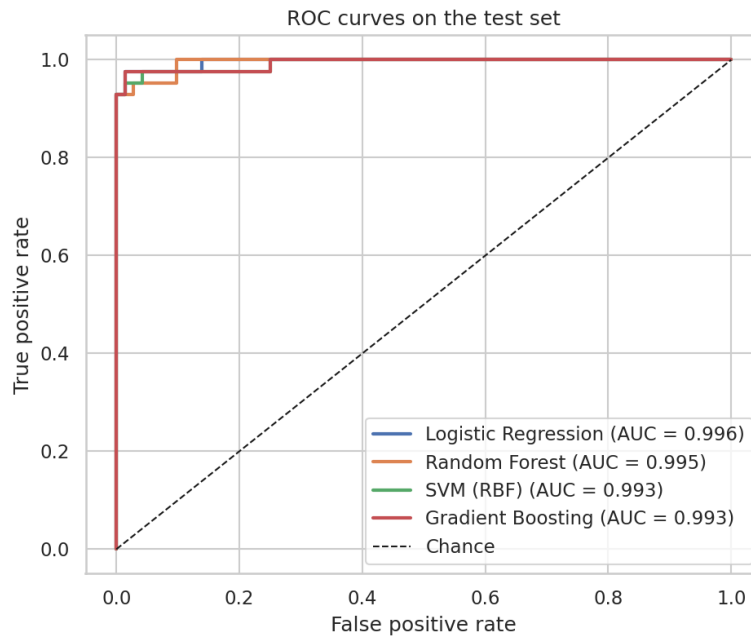


Figure 6. ROC curves on the test set. All models have AUC above 0.99.

5. Model selection and discussion

Selected: SVM (RBF) ($C = 10$, $\gamma = \text{scale}$). It had the best cross-validated F1 after tuning (0.967) and on the test set achieved:

	Score
Accuracy	0.974
Precision	1.000
Recall	0.929
F1	0.963
ROC-AUC	0.993

- **Precision 1.000:** every tumour flagged as malignant was truly malignant (0 false positives).
- **Recall 0.929:** 3 of 42 malignant tumours were missed. In a medical setting a missed cancer is the costlier error, so recall is the metric to improve, for example by lowering the decision threshold at the cost of some false positives.
- **Differences between models are small.** Cross-validated F1 spans only about 0.95 to 0.97 after tuning, and with 114 test samples a single extra mistake changes accuracy by 0.9 points. Logistic Regression is almost as good and far easier to explain, so it is a reasonable choice if interpretability matters more than the last fraction of a point.
- **No sign of overfitting:** cross-validation scores (F1 around 0.96) and test scores (F1 0.963) agree closely.

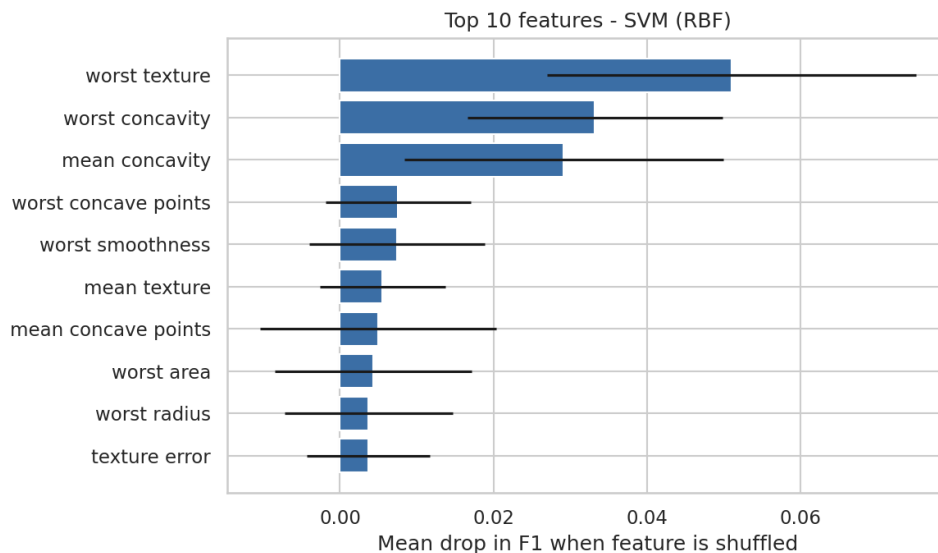


Figure 7. Permutation importance of the SVM on the test set. Texture and concavity features matter most. Because many features are strongly correlated, importance is shared among them and the error bars are wide.

6. Limitations and next steps

- Small, clean research dataset (569 samples); the results will not transfer directly to real clinical data.
- The 114-sample test set gives wide confidence intervals; repeated cross-validation or a larger dataset would give a firmer ranking.
- Next steps: threshold tuning to raise recall, probability calibration, and SHAP explanations for individual predictions.

7. How to reproduce (exact commands)

Repository: <https://github.com//alfido-task1-ml-classification>

Setup (Windows)

```
python -m venv .env
.venv\Scripts\activate
pip install -r requirements.txt
```

Setup (macOS / Linux)

```
python3 -m venv .env
source .env/bin/activate
pip install -r requirements.txt
```

Run

```
jupyter notebook notebooks/task1_ml_classification.ipynb # then Kernel > Restart & Run All
# or, headless, from the repo root:
python src/train_classification.py
```

Outputs: plots in `images/`, metrics in `results/`, saved model in `models/best_model.joblib`.

Deliverables: (1) notebook with code, plots and metrics, (2) this short report on model selection and results.

Deliverable checklist

- Data preprocessing, train/test split, cross-validation: done (sections 2 and 3)
- At least two algorithms compared: done (four algorithms)
- Accuracy, precision, recall, F1, ROC-AUC reported: done (section 4)
- Notebook with code, plots and metrics: `notebooks/task1_ml_classification.ipynb`
- README with environment setup and run commands: `README.md`